Last updated in October 2026

Waruni Kekulandara

Kandy, Sri Lanka | warunilkekulandara20@gmail.com | +94 77 158 4874

Personal Website | LinkedIn | GitHub

Research Interests

- Human-Computer Interaction
- Explainable and Interpretable AI
- Trust in Intelligent Systems
- Human-Centred Artificial Intelligence

Education

University of Peradeniya, B.Sc. Honours in Computer Science 2021 – 2025
Minor in Chemistry and Biology

- GPA: 3.80 / 4.00 (Top of the Batch | First-Class Honours)
- **Relevant Coursework:** Machine Learning, Neural Networks and Deep Learning, Object Oriented Analysis and Design, Software Engineering, Computer Graphics, Artificial Intelligence

Publications

Published (Peer-Reviewed Conference Papers)

- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2026). *Framework for Existing Counterfactual Methods with Feasibility and LLM Explanation Support.* In ***Proceedings of the 11th International Congress on Information and Communication Technology (ICICT 2026)***, DOI: 10.1007/978-3-032-35982-7_20.
- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2025). *Towards Counterfactual Verification for Tree Ensembles.* **2025 IEEE International Conference on Emerging Trends in Engineering and Computing (ETECOM 2025).** DOI: 10.1109/ETECOM66111.2025.11318990

Accepted

- **Kekulandara, W.**, De Silva, C., & Dayaratna, E. (2026). *Shades of Sentiment: Evaluating Emotional Fairness Across Ethnic Identities in LLMs.* **World Conference on Information Systems for Business Management (ISBM 2026)**, Bangkok, Thailand.

Under Review

- De Silva, C., **Kekulandara, W.**, & Dayaratna, E. (2026). *Is More Explanation Better? A Multi-Dimensional Study of Combined XAI Techniques for Regression.* **8th IEEE International Conference on Advancements in Computing (ICAC 2026)**, Colombo, Sri Lanka.

Conference Abstracts

- **Kekulandara, W.**, Amarasinghe, K., & Dayaratna, E. (2025). *A Unified Framework for Evaluating Existing Counterfactual Methods with Feasibility Constraints and LLM-Based Explanations.* **Sri Lanka Japan Collaborative Research (SLJCR 2025).**

Presentations

- **Shades of Sentiment: Evaluating Emotional Fairness Across Ethnic Identities in LLMs.** Research Cafe, University of Peradeniya (Poster Presentation), 2026 & ISBM 2026 (Virtual)
- **Framework for Existing Counterfactual Methods with Feasibility and LLM Explanation Support.** ICICT 2026 (Virtual).
- **Towards Counterfactual Verification for Tree Ensembles.** IEEE ETECOM 2025 (Virtual).
- **A Unified Framework for Evaluating Existing Counterfactual Methods with Feasibility Constraints and LLM-Based Explanations.** Sri Lanka Japan Collaborative Research (SLJCR 2025) (Virtual).

Awards & Honors

Alumni Prize for Excellence in Computer Science, University of Peradeniya 2026

- Awarded for achieving the highest GPA and graduating top of the Special Degree in Computer Science with an 'A' in the Research course.

University Award for Academic Excellence, University of Peradeniya 2026

- Recognized for maintaining outstanding academic performance.

Mensa International Qualification 2024

- Achieved a score within the top 2% of the general population on the proctored membership evaluation exam, fulfilling the official criteria for international society membership.

Research Experience

Final Year Research Project, University of Peradeniya 2024 – 2025

Enhancing Counterfactual Extraction Methods for Tree Ensembles

- Designed and implemented an evaluation framework for counterfactual explanation methods.
- Evaluated five selected counterfactual methods (Foil Trees, SHAP-C, LIME-C, LORE, CLEAR).
- Implemented an interactive Python UI incorporating Llama for visualizing counterfactual results and found that it explained infeasible counterfactuals just as fluently as feasible ones.
- Identified a design-level trade-off in existing counterfactual methods: correctness and feasibility need to be considered during generation, not enforced after it.

EEG Correlations of Creativity, Personality, and Vocational Interest: A Scoping Review, University of Peradeniya Ongoing

- Conducting an EEG experiment with university students to build a novel EEG dataset.
- Co-authoring an accompanying review paper as part of the same project.

LLM Agent for Emotional Fairness Across Ethnic Identities, University of Peradeniya Ongoing

- Extending the *Shades of Sentiment* study (accepted, ISBM 2026) by developing an autonomous agent framework to evaluate and mitigate emotional fairness across ethnic identities in LLMs.

Ethical and Technological Governance Framework Against coordinated harassment of celebrities in Digital Ecosystems, University of Peradeniya Ongoing

- Investigating platform moderation and policy approaches to online harassment targeting K-pop idols and other online public figures.

Teaching Experience

Demonstrator, Faculty of Science, University of Peradeniya Jan 2026 – Present

- Conducted practical laboratory sessions in Programming Laboratory II (Java), Computing for Science (Python), and Computer Graphics Programming (C++/OpenGL).
- Delivered lectures and facilitated academic discussions for theory courses, including Computer Architecture, Introduction to Computer Networks, Object-Oriented Programming, and Digital Image Processing.
- Developed practical assignments and examinations and assessed coursework and final examinations of 500+ total students.

Workshop Instructor, Postgraduate Institute of Science, University of Peradeniya

- Delivered a workshop for Master's students on explainable AI, covering SHAP, LIME, and counterfactual explanation methods.

Workshop Facilitator, Kingswood College, Kandy

- Facilitated a hands-on workshop introducing IoT concepts to school students.

Professional Experience

Project Manager, NoLoopTech (Pvt) Ltd Jun 2025 – Jul 2026

- Coordinated the development of 4 software products, managing project requirements, priorities, timelines, and deliverables from planning through deployment.
- Bridged communication between clients, management, developers, and designers to translate business goals and user needs into clear, actionable software requirements.
- Led requirements gathering, sprint planning, and task coordination for AI-powered and web application projects, aligning development priorities with stakeholder expectations and intended user experiences.
- Tracked project progress, documented decisions and requirements, and followed up on tasks and issues to maintain team alignment and keep deliverables on schedule.

Intern, Innovator Park International (Pvt) Ltd Oct 2023 – Feb 2024

- Actively participated in the STREAM Bootcamp program, demonstrating a deep understanding of course development and content design.
- Delivered engaging and impactful educational lectures to program participants (IOT, Biology, Chemistry).
- Spearheaded the creation and execution of the Peradeniya Hospital Innovation Workshop for Doctors.
- Built a collaborative environment designed to help medical professionals explore novel solutions and practical applications.

Selected Additional Projects

- **Health-Mate:** Full-stack web application designed and built end-to-end (React, TypeScript, Java, PostgreSQL).
- **eBook Hosting Website:** PHP-based platform for hosting student-authored eBooks, with front-end and database design.
- **Sydney Opera House 3D Model:** Detailed 3D reconstruction using OpenGL, C++, GLUT, and SOIL.

Technical Skills

- **Programming Languages:** Python, Java, C, C++, JavaScript
- **AI/ML:** NumPy, Pandas, Scikit-learn, Matplotlib, PyTorch, TensorFlow, SHAP, LIME
- **Databases:** MySQL, PostgreSQL
- **Tools:** Git/GitHub, Visual Studio Code, IntelliJ, Google Colab
- **Frameworks & Web:** React, Spring Boot, RESTful APIs, HTML, CSS

Professional Organizations and Affiliations

- **Editor**, Astronomical Society, University of Peradeniya (2025 – 2026)
- **Program Team Member**, IEEE Computer Society, University of Peradeniya (2024 – 2025)
- **Junior Treasurer**, Drama Society, University of Peradeniya (2023 – 2024)
- **President Guide**, Sri Lanka Girl Guides Association (2013 – 2019)

Volunteering

- RESCON, Postgraduate Institute of Science, University of Peradeniya (2026)
- Huddle Workshop, University of Peradeniya (2026, 2024, 2023, 2022)
- Project for Special Needs Children, Blue Rose School (2023)
- Save the Next Generation - Knuckles, University of Peradeniya (2022)

Languages

English (IELTS:8) | Sinhala (Native) | Tamil (Reading) | Korean (Reading)